NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

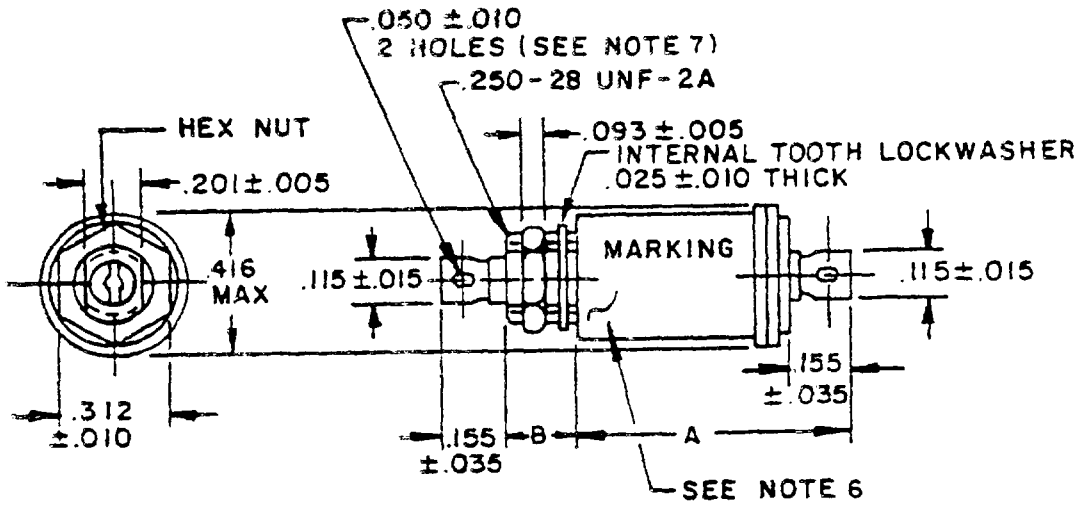
MIL-PRF-15733/23A
1 February 1982
SUPERSEDING
MIL-F-15733/23
7 July 1971

PERFORMANCE SPECIFICATION SHEET

FILTERS, RADIO FREQUENCY INTERFERENCE, HERMETICALLY SEALED, STYLES FL10 AND FL11

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the filters described herein shall consist of this specification and the latest issue of MIL-F-15733.

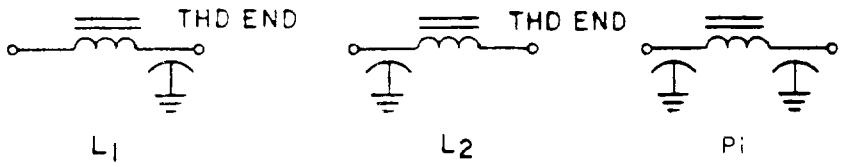


DIMENSION					
A = .630 inches max				A = .730 inches max	
B ±.010				B ±.010	
.312		.187		.312	.187
Dash no.		Dash no.		Dash no.	Dash no.
0001	0031	0002	0032	0005	0006
0003	0033	0004	0034	0011	0012
0007	0037	0008	0038	0017	0018
0009	0039	0010	0040	0023	0024
0013	0043	0014	0044	0029	0030
0015	0045	0016	0046	0035	0036
0019	0049	0020	0050	0041	0042
0021	0051	0022	0052	0047	0048
0025	0055	0026	0056	0053	0054
0027	0057	0028	0058	0059	0060

FIGURE 1. Case dimensions and circuit diagrams

FSC 5915

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CIRCUIT DIAGRAMS

INCH	MM	INCH	MM
.005	.13	.155	3.94
.010	.25	.187	4.75
.015	.38	.201	5.11
.025	.64	.250	6.35
.035	.89	.312	7.92
.050	1.27	.416	10.57
.070	1.78	.630	16.00
.093	2.36	.730	18.54
.115	2.92		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. Circuit diagrams are for information only.
4. All filters shall be supplied with mounting hardware.
5. Use of styles FL10 or FL11 with or without shoulder is optional.
6. Terminal identification (non-symmetrical filters): The case shall be marked at the threaded end of the filter with the symbol "C" or "L" as follows:

Symbol	Circuit
C - - - - - - - - -	L_1
L - - - - - - - - -	L_2

7. Optional terminal hole of .070 +.010 inch or terminal slot of .050 +.010 x .070 +.010 inch may be supplied.

FIGURE 1. Case dimensions and circuit diagrams - Continued.

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TABLE I. Electrical characteristics.

Dash number	Circuit diagram	Max rated current (amperes)	Rated dc voltage (volts)	Max voltage drop (volts)	Minimum Insertion Loss (dB) in accordance with MIL-STD-220 1/																						
					At 25°C											At -55°C and +125°C											
					15 kHz	30 kHz	50 kHz	100 kHz	150 kHz	300 kHz	500 kHz	1 MHz	10 MHz	1 GHz	1 GHz	15 kHz	30 kHz	50 kHz	100 kHz	150 kHz	300 kHz	500 kHz	1 MHz	10 MHz	1 GHz	
0001	L_2	.10	50	.17	9	20	29	41	48	60	69	70	70	70		7	18	27	39	46	58	67	70	70	70	
0002	L_2																									
0003	L_1																									
0004	L_1																									
0005	Pi				15	36	50	69	79	80	80	80	80	80		13	34	48	67	77	78	80	80	80	80	
0006	Pi				15	36	50	69	79	80	80	80	80	80		13	34	48	67	77	78	80	80	80	80	
0007	L_2		150		2	11	20	32	39	51	60	60	60	60		-	9	18	30	37	49	58	60	60	60	
0008	L_2																									
0009	L_1															-										
0010	L_1															-										
0011	Pi				-	17	31	49	60	70	70	70	70	70		-	15	29	47	58	70	70	70	70	70	
0012	Pi				-	17	31	49	60	70	70					-	15	29	47	58	70	70				
0013	L_2	.30	50	.23	6	15	23	35	42	54	63					4	13	21	33	40	52	61				
0014	L_2																									
0015	L_1																									
0016	L_1																									
0017	Pi				4	29	44	62	73	80	80	80	80	80		2	27	42	60	71	80	80	80	80	80	
0018	Pi				4	29	44	62	73	80	80	80	80	80		2	27	42	60	71	80	80	80	80	80	
0019	L_2		150		-	6	13	25	32	44	52	60	60	60		-	4	11	23	30	42	50	60	60	60	
0020	L_2				-											-										
0021	L_1				-											-										
0022	L_1				-											-										
0023	Pi				-	8	24	43	53	70	70	70	70	70		-	6	22	41	51	69	70	70	70	70	
0024	Pi				-	8	24	43	53	70	70	70				-	6	22	41	51	69	70	70			
0025	L_2	.50	50	.18	5	12	19	29	36	48	57	69				3	10	17	27	34	46	55	67			
0026	L_2																									
0027	L_1																									
0028	L_1																									
0029	Pi				-	21	37	56	67	80	80	80	80	80		-	19	35	54	65	78	80	80	80	80	
0030	Pi				-	21	37	56	67	80	80	80	80	80		-	19	35	54	65	78	80	80	80	80	

see footnote at end of table.

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TABLE I. Electrical characteristics - Continued.

Dash number	Circuit diagram	Max rated current (amperes)	Rated dc voltage (volts)	Max voltage drop (volts)	Minimum insertion loss (dB) in accordance with MIL-STD-220 1/																							
					At 25°C											At -55°C and +125°C												
					15 kHz	30 kHz	50 kHz	100 kHz	150 kHz	300 kHz	500 kHz	1 MHz	10 MHz	1 GHz		15 kHz	30 kHz	50 kHz	100 kHz	150 kHz	300 kHz	500 kHz	1 MHz	10 MHz	1 GHz				
0031	L_2	.50	150	.18	-	3	9	20	26	39	47	59	60	60		-	1	7	18	24	37	45	57	60	60			
0032	L_2				-											-												
0033	L_1				-											-												
0034	L_1				-											-												
0035	Pi				-	-	15	37	48	66	70	70	70	70		-	-	13	35	46	64	68	70	70	70			
0036	Pi				-	-		37	48	66	70	70				-	-		35	46	64	68	70					
0037	L_2	1.0	50	.14	5	11		21	26	36	44	55				3	9		19	24	34	42	53					
0038	L_2																											
0039	L_1																											
0040	L_1																											
0041	Pi				-	-	20	41	52	71	80	80	80	80		-	-	18	39	50	69	78	80	80	80			
0042	Pi				-	-	20	41	52	71	80	80	80	80		-	-	18	39	50	69	78	80	80	80			
0043	L_2		150		-	3	6	12	16	26	34	46	60	60		-	1	4	10	14	24	32	44	60	60			
0044	L_2				-											-												
0045	L_1				-											-												
0046	L_1				-											-												
0047	Pi				-	-	-	18	32	51	65	70	70	70		-	-		16	30	49	63	68	70	70			
0048	Pi				-	-	-	18	32	51	65	70				-	-	-	16	30	49	63	68					
0049	L_2	2.0	50		5	10	14	20	24	32	38	48				3	8	12	18	22	30	36	46					
0050	L_2																											
0051	L_1																											
0052	L_1																											
0053	Pi				-	-	-	33	46	65	76	80	80	80		-	-	-	31	44	63	74	80	80	80			
0054	Pi				-	-	-	33	46	65	76	80	80	80		-	-	-	31	44	63	74	80	80	80			
0055	L_2		150		-	3	5	11	15	23	29	38	60	60		-	1	3	9	13	21	27	36	60	60			
0056	L_2				-											-												
0057	L_1				-											-												
0058	L_1				-											-												
0059	Pi				-	-	-	-	22	45	59	70	70	70		-	-	-	-	20	43	57	70	70	70			
0060	Pi				-	-	-	-	22	45	59	70	70	70		-	-	-	-	20	43	57	70	70	70			

1/ Full load insertion loss measurements shall be performed at frequencies between 100 kHz to 10 MHz inclusive; all other measurements shall be performed at no-load.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case: Metal.

Terminals: Solder lug.

Operating temperature range: -55° to +125°C.

Rated voltage: See table I.

Rated current: See table I.

Insertion-loss characteristic: W (see table I).

Creepage and clearance: Not applicable.

Seal: Method 112, test condition A, MIL-STD-202.

Capacitance to ground: Not applicable.

Temperature rise: 25°C, maximum.

Dielectric withstanding voltage: Method 301, MIL-STD-202.

Test temperature: 25°C.

Test voltage: 2.5 times the rated voltage, applied between each terminal and ground, for a period of 1 to 5 seconds.

Maximum charging current: 50 milliamperes.

Barometric pressure (reduced): Method 105, test condition B (50,000 ft), MIL-STD-202.

Dielectric withstanding voltage: 125 percent of the rated voltage.

Maximum charging current: 50 milliamperes.

Insulation resistance: Method 302, MIL-STD-202.

Test temperature: 25°C.

Test potential: Rated dc voltage.

Maximum charging current: 50 milliamperes.

Insulation resistance: Shall be not less than 1,000 megohms.

Voltage drop: See table I.

Insertion loss: In accordance with MIL-STD-220 and table I.

Insertion loss (check test): In accordance with MIL-F-15733.

Overload: Filters shall be suspended by their terminals in free air at an ambient temperature of 25° ±5°C. A test current of 140 percent of rated current shall be applied for 15 minutes. After the filter has returned to room temperature, the insulation resistance and voltage drop measurements shall meet the initial requirements.

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Terminal strength: Method 211, test condition A (pull), MIL-STD-202.

Force: 5 pounds.

Resistance to soldering heat: Method 210, test condition B, MIL-STD-202.

Salt spray (corrosion): Method 101, test condition B, MIL-STD-202.

Thermal shock and immersion:

Thermal shock: Method 107, test condition A, MIL-STD-202, except that the temperature at step 3 shall be +125°C.

Immersion: Method 104, test condition A, MIL-STD-202.

Measurements after final cycling:

Dielectric withstanding voltage: 90 percent of the initial voltage shall be applied for 5 ±1 seconds.

Maximum charging current: 50 milliamperes.

Insulation resistance: Shall be not less than 30 percent of the initial requirement.

Insertion loss (check test): Shall meet the initial requirements.

Shock (specified pulse): Method 213, test condition I, MIL-STD-202.

Electrical load: 100 percent of rated voltage shall be applied between terminals and case.

Vibration, high frequency: Method 204, test condition D (20g), MIL-STD-202.

Electrical load: Rated voltage and current shall be applied continuously during test.

Test during vibration: During the last cycle in each direction, an electrical measurement shall be made to determine intermittent open- or short-circuiting.

After vibration test: Filters shall be subjected to and shall meet the initial requirements of the insertion loss (check test).

Moisture resistance: Method 107, test condition A, MIL-STD-202, except that the temperature of step 3 shall be +125°C. No measurements shall be made before and after cycling. Filters shall then be tested in accordance with method 106, MIL-STD-202.

Initial measurements: Not applicable.

Polarization voltage: 100 volts dc or rated voltage, whichever is less.

Measurements after 24-hour drying period:

Dielectric withstanding voltage: 90 percent of the initial voltage shall be applied for 5 ±1 seconds.

Maximum charging current: 50 milliamperes.

Insulation resistance: Shall meet the initial requirement.

Insertion loss (check test): Shall meet the initial requirements.

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Life: Method 108, MIL-STD-202, test condition D (1,000 hr) for qualification inspection; test condition B (250 hr) for group C inspection.

Test temperature: +125°C.

Test voltage: 1.4 times rated dc voltage.

Measurements after test:

Dielectric withstanding voltage: 90 percent of the initial voltage shall be applied for 5 ±1 seconds.

Maximum charging current: 50 milliamperes.

Insulation resistance: Shall be not less than 30 percent of the initial requirement.

Insertion loss (check test): Shall meet the initial requirements.

Part number: M15733/23-(dash number from table I).

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
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Navy - EC
Air Force - 85

Review activities:
Army - AR, MI
Navy - OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AT, AV, ME
Navy - AS, MC, SH
Air Force - 19

Agent:
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Preparing activity:
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